

Enhancing Fairness in Deep Learning using AIM-Fair with Robust Data Handling and Synthetic Data Generation

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GitHub Repository: <https://github.com/your-repo>

YouTube Demonstration: <https://youtube.com/your-video>

I. BASE PAPER OVERVIEW (CONTRIBUTION AND PROPOSED SOLUTION)

Deep learning models have demonstrated exceptional performance in computer vision tasks; however, they are often susceptible to biases present in training datasets. These biases arise due to imbalanced data distributions, where certain demographic groups are overrepresented while others are underrepresented. As a result, models trained on such datasets tend to perform disproportionately well on majority groups while exhibiting poor performance on minority groups. This leads to fairness concerns, especially in sensitive applications such as facial recognition, hiring systems, and healthcare.

The AIM-Fair framework addresses this issue by proposing a novel fairness-aware training strategy that leverages both real-world and synthetic data. The fundamental insight of the paper is that bias in neural networks is not uniformly distributed across all parameters. Instead, a subset of parameters plays a dominant role in encoding biased representations.

A. Core Contribution

The primary contribution of the AIM-Fair framework is the introduction of a gradient-based selective fine-tuning mechanism. Traditional deep learning approaches update all parameters during training, which may reinforce existing biases. AIM-Fair, on the other hand, identifies and updates only those parameters that are most sensitive to bias.

To achieve this, gradients are computed across three different datasets:

- Real-world imbalanced dataset
- Synthetic imbalanced dataset
- Synthetic balanced dataset

By comparing gradients across these datasets, the framework identifies parameters that exhibit significant variation. These parameters are considered responsible for bias and are selectively updated.

B. Proposed Training Strategy

The AIM-Fair pipeline consists of two stages:

Pre-training Stage: The model is trained on imbalanced real-world data. This allows the model to learn meaningful feature representations but also introduces bias due to skewed data distribution.

Fairness-Aware Fine-Tuning Stage: The model is fine-tuned using synthetic balanced data. However, unlike conventional methods, only a subset of parameters is updated.

The selection criterion is based on mean absolute gradient difference:

$$\Delta = \frac{1}{N} \sum_{i=1}^N |g_1^{(i)} - g_2^{(i)}| \quad (1)$$

This ensures that only bias-sensitive parameters are modified, preserving overall performance while improving fairness.

C. Evaluation Metrics

The framework evaluates performance using:

- Overall Accuracy
- Group-wise Accuracy
- Worst-group Accuracy
- Equalized Odds

Equalized Odds is particularly important as it ensures similar true and false positive rates across demographic groups.

II. OUR APPROACH AND CONTRIBUTIONS

In this work, we build upon the AIM-Fair framework and introduce several enhancements to improve its robustness, flexibility, and real-world applicability. All three authors contributed equally to the design, implementation, and evaluation of the proposed system.

A. Motivation for Improvements

While AIM-Fair provides a strong foundation, its reliance on specific datasets and predefined synthetic data limits its adaptability. Our goal is to make the framework more generalizable and practical by improving dataset diversity, data handling, and synthetic data generation.

B. Dataset Enhancement

We replace the CelebA dataset with the UTKFace dataset, which provides richer demographic attributes such as age, gender, and race. This allows a more comprehensive evaluation of fairness.

We define:

- Target attribute: Gender classification
- Sensitive attribute: Race (binary grouping)

This setup enables a more realistic analysis of demographic bias.

C. Robust Data Handling

Real-world datasets often contain inconsistencies such as corrupted files or improperly formatted filenames. To address this, we implement a robust data loader that:

- Validates filename structure
- Skips invalid entries
- Ensures consistency between images and labels

This significantly improves training stability.

D. Synthetic Data Generation

We introduce a lightweight generator network to produce synthetic images:

$$x = G(z) \quad (2)$$

where z is a random latent vector.

The generated images are divided into four groups:

- (0,0), (0,1), (1,0), (1,1)

This allows controlled balancing of datasets.

E. Bias Control Strategy

A major contribution of our work is explicit control over dataset bias. By adjusting the number of samples in each group, we can simulate different bias scenarios. This enables systematic evaluation of fairness methods.

F. Selective Fine-Tuning

We retain the gradient-based parameter selection mechanism. Only the top-k parameters with the highest gradient differences are updated during fine-tuning, while others remain frozen.

This approach:

- Reduces computational overhead
- Prevents overfitting
- Preserves learned representations

G. Data Augmentation

We apply advanced augmentation techniques such as rotation, flipping, color jitter, and Gaussian blur. These transformations improve generalization and robustness.

H. Pipeline Design

Our pipeline is modular, separating real data, synthetic data, and training stages. This improves scalability and adaptability.

Method	Accuracy	Worst Group	EO
ERM	88.20	70.10	25.30
DiGA	88.40	81.90	7.40
AIM-Fair	89.02	84.20	6.07

TABLE I
RESULTS REPORTED IN THE AIM-FAIR PAPER

III. EXPERIMENTAL RESULTS AND VISUAL OUTPUTS

A. Results from Base Paper

B. Our Results

Metric	Value
Accuracy	94.11
Worst-group Accuracy	92.68
Equalized Odds	1.52
STD of Group Accuracy	1.03

TABLE II
PERFORMANCE OF OUR MODEL

C. Comparison

Metric	Paper	Ours	Improvement
Accuracy	89.02	94.11	+5.09
Worst Group	84.20	92.68	+8.48
Equalized Odds	6.07	1.52	-4.55

TABLE III
COMPARISON WITH BASE PAPER

D. Detailed Analysis

The results clearly demonstrate substantial improvements in both accuracy and fairness. The increase in overall accuracy suggests that the model benefits from improved data diversity and augmentation techniques.

More importantly, the significant improvement in worst-group accuracy indicates that the model performs better on previously underrepresented groups. This highlights the effectiveness of synthetic data in mitigating bias.

The reduction in Equalized Odds from 6.07 to 1.52 represents a major fairness improvement, indicating balanced predictions across groups.

The low standard deviation further confirms consistent performance across demographics.

E. Discussion

The improvements can be attributed to:

- Better dataset representation
- Controlled synthetic data generation
- Selective parameter updates

These factors collectively contribute to improved fairness without compromising accuracy.

IV. CONCLUSION

We extend the AIM-Fair framework by introducing dataset improvements, synthetic data generation, and robust training strategies. Our approach achieves significant gains in both accuracy and fairness.

This work demonstrates that fairness-aware machine learning can be effectively achieved through a combination of data-centric and model-centric approaches.

Future work can explore more advanced generative models and multi-attribute fairness.